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Causal Concept Explanations for Deep Neural Models

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Abstract

Concept-based explanations in neural models are important for the development, certification, and adoption of autonomous systems. Yet, interpretability remains correlational. Without methods to guide and quantify causal relationships between explanations, a concept explanation can be provided without being necessary or sufficient for the observed output thereby appearing correct while misleading. We introduce Causal Concept Wrapper Network (CCW-Net), the first use of interventional mediation analysis from fields such as economics, medicine, and epidemiology as a training objective to align causal effects of concept-based explanations with a model's output. We further apply mediation analysis as an evaluation tool to quantify, per concept label combination, how much of any model's output is causally attributable to each concept explanation. As a training objective, CCW-Net shapes a model's concept representations such that they are both necessary and sufficient for the outputs they are credited with and independent of other concepts' contributions. Counterfactual concept representations anchor a set of causal alignment criteria: necessity, sufficiency, and independence. Necessity drives irrelevant concept effects toward zero, sufficiency ensures relevant concept representations fully reconstruct a model's output, and independence enforces the total effect decomposes linearly across concepts. We evaluate CCW-Net across aircraft control, driving, and image classification. CCW-Net provides a step toward concept-based explanations that are not merely interpretable, but quantifiably causal, supporting safer evaluation, certification, and deployment of transparent and trustworthy neural models.

1 Introduction

Learned policies in autonomous systems are moving from research labs into high-stakes, real-world settings such as aviation [Pope et al., 2023, Ward, 2023], driving [Phan-Minh et al., 2023], and robotics [Tang et al., 2025]. This is also the case for perception models in safety-relevant settings such as medical imaging. In task and safety-critical domains such as these, interpretability is becoming a necessary component [Rudin, 2019, Atakishiyev et al., 2025]. Developers need it to troubleshoot [Kenny et al., 2026], testers need it to verify [Rountree et al., 2021, Mahmud et al., 2024], and users need it to understand [Sanneman and Shah, 2022]. Yet many neural models remain black boxes: they are not interpretable in a way that is useful for people in the task at hand.

Concept-based interpretability offers promise by structuring decisions through human-interpretable concepts [Koh et al., 2020, Echterhoff et al., 2024]. However, the link between concepts and outputs is typically correlational, not causal. A concept can activate alongside an output without being in the causal chain [Zhou et al., 2022]. An autonomous vehicle policy might activate a "passenger pickup" concept that has no causal bearing on the observed action. A bird classifier may activate

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"has red plumage" on every image of a cardinal while its prediction is actually driven by "has crest". The explanation appears correct but may mislead. To be useful for troubleshooting, certifying, and auditing, explanations must capture causal effects, i.e., the contribution of each concept to the output, isolated from concepts it co-occurs with [Pearl, 2012]. Correlation, no matter how sophisticated, does not address such causal questions [Pearl, 2009]. For safety and task-critical deployment we must transition from correlation to causation.

Beyond quantifying causal effects, some concept representations themselves limit what explanations can convey. Prior work typically represents concepts as scalars [Madumal et al., 2020, Koh et al., 2020, Kenny et al., 2026], e.g., a "left turn" in driving or a "has yellow belly" attribute in bird classification. Scalars indicate the presence or strength of a concept but cannot capture their higher-dimensional, context-dependent expressions. We extend concept representations from scalars to vectors, enabling each concept to encode both *whether* it exists and *how* it is expressed.

We address these challenges with Causal Concept Wrapper Network (CCW-Net), the first use of mediation analysis as a training objective in deep networks. CCW-Net introduces a network architecture and adapts interventional multi-mediator mediation analysis from economics and epidemiology [Vansteelandt and Daniel, 2017, Loh et al., 2022, Celli, 2022, Lee et al., 2021] to shape a model's concept representations so that each concept's contribution to the output is locally necessary, sufficient, and independent of contributions from other concepts. Counterfactuals required by these criteria are drawn from a mediator model (in our experiments, a per-concept invertible flow) trained alongside the overall model to track the conditional distribution of its concept embeddings. We further apply mediation analysis as an evaluation tool, quantifying, for each concept label combination, the share of a model's output causally attributable to each concept. This is finer-grained than the population-level alignment scores reported in prior concept-based work. We evaluate CCW-Net across aircraft control, driving, and fine-grained image classification, demonstrating that mediation-trained concepts produce explanations that are not merely interpretable, but quantifiably causal.

Main contributions. Our contributions are threefold:

1. **Training objective for causal concepts in deep networks:** An interventional mediation analysis based objective that shapes each concept's causal effect on a model's output so that activated concepts are those causing the output thereby improving explanation trustworthiness.
2. **Network architecture for concept-level mediation analysis:** A network architecture that drives absent concept representations toward zero supporting well-defined interventional mediation analyses and related training objectives (App. C.1).
3. **Evaluation method for concept-based causality claims:** For any concept-bottleneck model, a method to quantify, for each concept label combination, how much a change in the model's output is causally attributable to each concept (App. F).

2 Related work

Concept-based explanations. Concept-based interpretability is widely explored [Kim et al., 2018, Alvarez-Melis and Jaakkola, 2018, Koh et al., 2020, Bai et al., 2023, Achtibat et al., 2023, Tan et al., 2024]. Concept Bottleneck Models [Koh et al., 2020] and adaptations to sequential decision-making [Kenny et al., 2023, 2026, Delfosse et al., 2024, Das et al., 2023] structure decisions through human-interpretable concepts but typically lack formal causal analysis.

Causal mediation analysis. Causal mediation analysis models systems as directed acyclic graphs and decomposes an input's effect on an outcome into pathways through intermediate variables called mediators. Foundational tools have been developed in economics, medicine, and epidemiology [MacKinnon, 2008, Imai et al., 2010, VanderWeele, 2015, Pearl, 2012, Celli, 2022, Lee et al., 2021]. Multi-mediator extensions [Vansteelandt and Daniel, 2017, Loh et al., 2022] decompose total effects across multiple mediators without requiring causal ordering or independence assumptions among them, with indirect effects quantifying how much of an effect operates through a specific mediator. These tools are designed to estimate causal effects from observational data under strict assumptions (App. D), but to date have been applied to analyze causal relationships in data collected outside randomized controlled trials or trained models post-hoc, not as training objectives that shape deep networks' representations during learning.

Causal and disentanglement-based explanations for deep networks. A growing body of work applies causal frameworks to trained deep networks. Vig et al. [2020] applied mediation analysis post hoc to study bias in transformer language models, treating individual neurons and attention heads as mediators. Geiger et al. [2021, 2025] formalize causal abstraction as a way to validate that a trained network implements a higher-level causal model. In concept-based interpretability specifically, recent work imposes or discovers causal structure over learned concepts [De Felice et al., 2025, Dominici et al., 2025a, Qian et al., 2024]. Disentanglement methods like β-TCVAE [Chen et al., 2018] and iVAE [Khemakhem et al., 2020] learn factorized representations evaluated by DCI [Eastwood and Williams, 2018] and FactorVAE score [Kim and Mnih, 2018]. Across these methods, causal or representational analysis is applied post hoc to a model trained with conventional objectives, and disentanglement is a representational property that need not imply causal alignment [Schölkopf et al., 2021]. CCW-Net differs in that causal relationships between concept representations and model outputs are shaped during training rather than audited post-hoc.

Feature attribution. Saliency-based [Simonyan et al., 2014, Selvaraju et al., 2017, Smilkov et al., 2017, Sundararajan et al., 2017] and perturbation-based [Ribeiro et al., 2016, Lundberg and Lee, 2017] attribution methods explain decisions by associating inputs or features to outputs. While useful for highlighting what features a model's outputs correspond to, these methods generally remain correlational. Ablating a feature changes a prediction without establishing causal responsibility [Adebayo et al., 2018, Zhou et al., 2022]. CCW-Net grounds causal attribution in interventional indirect effect estimates, yielding per-concept attributions with quantifiable causal estimates.

3 Problem formulation and preliminaries

We seek to train an interpretable policy that achieves expert-level task performance, provides concept-based explanations for its predicted outputs $\hat{Y}$, and ensures concept–output relationships reflect causal effects rather than spurious correlations. We assume access to a frozen pretrained backbone f that achieves good task performance but lacks interpretability, and a dataset $\mathcal{D} = \{(X_i, Y_i, c_i)\}_{i=1}^{N}$ where $X_i \in \mathcal{X}$ is an input, $Y_i \in \mathcal{Y}$ is the expert output, and $c_i \in \{0,1\}^J$ is a vector of J human-interpretable concept labels. We assume the concept ontology captures the key factors driving the model's outputs, the dataset provides sufficient ground truth, and standard ignorability conditions hold for the induced concept-to-output structural causal model (App. D). We jointly optimize task fidelity, concept correctness, and causal alignment (Sec. 4), with concept–output causal relationships quantified in evaluation.

Structural causal models. A structural causal model (SCM) [Pearl, 2009, Peters et al., 2017] consists of equations $X_i = k_i(\mathrm{pa}_i, u_i)$ where each equation represents a mechanism determining exactly one variable, with u_i the exogenous error term and pa_i the endogenous parent variables that directly cause X_i.

Interventional mediation analysis. Mediation analysis [Pearl, 2012] decomposes the causal effect of a treatment on an outcome Y into pathways that operate through variables called *mediators* M. For mediators $M = (M_1, \ldots, M_J)$, the total effect TE decomposes as $\mathrm{TE} = \mathrm{DE} + \mathrm{IE}$ into direct effect (DE) and indirect effect (IE) components. With multiple mediators, the indirect effect further decomposes per mediator j as $\mathrm{IE} = \sum_j \mathrm{IE}_j$. Following Loh et al. [2022], we relax the assumption of causal ordering or independence among mediators by introducing a residual interaction term IE_μ, yielding $\mathrm{TE} = \mathrm{DE} + \sum_j \mathrm{IE}_j + \mathrm{IE}_\mu$. This relaxation is important for deep networks where operations like softmax and cross-attention induce dependencies among concept activations, which IE_μ absorbs without violating identifiability. The *interventional indirect effect* IE_j [Vansteelandt and Daniel, 2017] quantifies how much of the total effect operates through mediator j when its representation M_j is set to a counterfactual value while other mediator representations are held fixed. These quantities are identifiable from observational data under standard ignorability conditions [VanderWeele, 2015, Vansteelandt and Daniel, 2017, Loh et al., 2022], which hold by construction in the wrapper SCM (App. D).

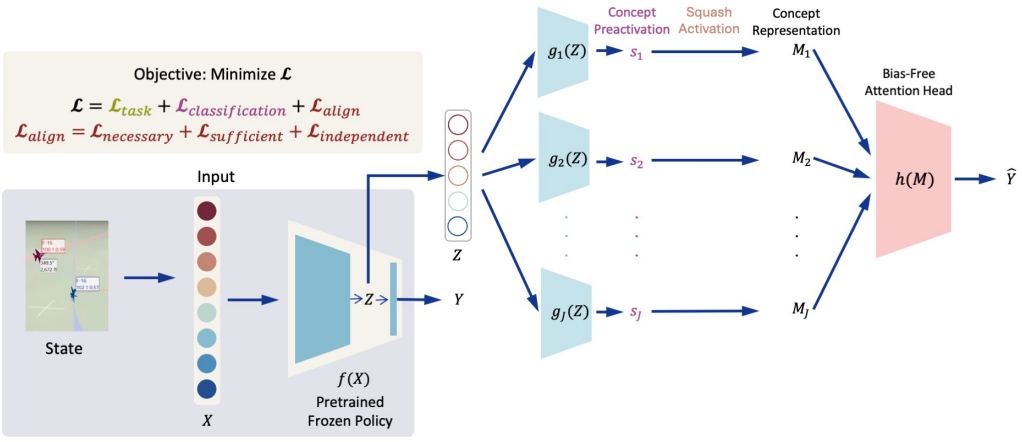

Figure 1: **CCW-Net architecture.** A concept encoder g_θ maps the frozen backbone's latent Z to one squashed vector M_j per concept, whose length encodes the concept's presence and whose direction encodes its context-dependent expression, and a bias-free attention head h_ϕ maps M to the wrapper output $\hat{Y}$. Causal alignment losses (weights and the auxiliary presence term omitted) are added to the task and concept losses, so that training shapes, and evaluation measures, how much each concept causally drives $\hat{Y}$.

4 Causal Concept Wrapper Network (CCW-Net)

CCW-Net wraps a frozen backbone policy with a concept encoder and new policy head which are trained to imitate the backbone's task performance as well as classify human-interpretable concepts similar to Kenny et al. [2026] but using a different architecture. In addition, training involves three causal alignment criteria to induce causal relationships between concepts and outputs: necessity, sufficiency, and independence. These criteria are then evaluated via interventional mediation analysis on the wrapper's outputs. Each causal alignment loss compares the head's output before and after one change of a training input's resulting concept j's representation. In mediation analysis terms, the concept representations are mediators and the policy head is the outcome model. Every effect discussed is the model's own concept-to-output effect within the wrapper architecture's SCM, not a claim about inputs or observations to output effects. For our experiments, counterfactual concept representations required by the necessity alignment term are drawn from a per-concept invertible flow trained on the wrapper's encoder concepts. Any suitable modeling approach may be used. Reported evaluation metrics are produced from a separate estimator that does not use the mediator model.

4.1 Architecture

Given a frozen pretrained backbone $f : \mathcal{X} \to \mathcal{Z}$ where $\mathcal{Z}$ is the frozen backbone's last latent layer, a concept encoder $g_\theta : \mathcal{Z} \to \mathbb{R}^{J \times D}$ maps $Z = f(X)$ to a concept representation $M = (M_1, \ldots, M_J)$. Each $M_j \in \mathbb{R}^D$ represents the j-th human-interpretable concept. An attention head $h_\phi : \mathbb{R}^{J \times D} \to \mathcal{Y}$ maps the concept representation to a wrapper output $\hat{Y} = h_\phi(M)$. The wrapper induces the SCM $X \to Z \to M \to \hat{Y}$ with no direct $Z \to \hat{Y}$ edge so the input reaches the output only through the concept representations M_j (DE=0), which gives mediator-outcome ignorability by construction (App. D, Prop. 2). The concept encoder g_θ is a multi-layer perceptron producing J concept representation vectors of dimension D, each passed through a squash nonlinearity [Sabour et al., 2017] mapping each slot length onto $[0, 1)$. The concept classification loss trains this length as presence leaving the direction free to encode the concept's context-based expression.

The concept classification loss drives inactive and active representations toward length zero and one, respectively. A zero concept vector representation carries a zero value vector in the attention head, so zeroing a concept removes its value contribution to $\hat{Y}$ regardless of its attention weight (Prop. 1).

4.2 Mediator model: Counterfactual concepts via a per-concept flow

Necessity asks whether the lack of a given concept's representation results in a lack of that concept manifesting in the model's outcome. In interventional mediation analysis, this is tested by replacing an inactive concept representation with an active one for the same input case. By definition, this is an unobservable counterfactual case. To overcome this, we reflect the approach of Vansteelandt and Daniel [2017] and draw counterfactual mediator expressions from a mediator model learned from observational data. In our experiments, we invoke a per-concept invertible flow $\mathcal{F}_\psi$ to model mediator distributions $p(M_j|c_j, X)$ and draw counterfactual examples where the concept label is flipped between present and not-present,

$$\tilde{M}_j \sim \mathcal{F}_\psi(\cdot \mid 1 - c_j, X). \tag{1}$$

We fit the flow by maximum likelihood to the encoder's mediator representations after a warmup (Sec. 4.4, App. G, Algorithm 1). Our method is not specific to flow modeling. Any modeling approach may be chosen that best serves the domain.

4.3 Causal alignment criteria

Each causal alignment criterion (necessity, sufficiency, independence) is stated in terms of a change in the head output caused by an edit of a concept vector. Let $\mathbf{0}$ be the all-zero concept vector, $\mathbf{0}_{j\leftarrow M_j}$ the all-zero vector with slot j set to its realized value M_j, and $M_{j\leftarrow\tilde{M}_j}$ the realized concepts M with slot j replaced by its counterfactual draw $\tilde{M}_j$ (Eq. 1). We define three changes:

$$\Delta^0 = \hat{Y} - h_\phi(\mathbf{0}), \qquad \Delta^0_j = h_\phi(\mathbf{0}_{j\leftarrow M_j}) - h_\phi(\mathbf{0}), \qquad \Delta^{\text{cf}}_j = h_\phi(M_{j\leftarrow\tilde{M}_j}) - \hat{Y}. \tag{2}$$

Δ^0 switches every concept slot on from $\mathbf{0}$ to its realized value, Δ^0_j switches slot j alone on from $\mathbf{0}$, and Δ^{cf}_j swaps slot j for its counterfactual draw, keeping all other slots at their realized values. The subscript indicates the edited slot, and the superscript indicates the edit: switching slots on from $\mathbf{0}$ (0) or swapping in a counterfactual draw from the mediator model (cf). In mediation analysis terms, the concept label flip in place of a treatment change gives Δ^{cf}_j as the per-sample analog of the interventional indirect effect IE_j of Sec. 3 and Δ^0_j the analogous contrast of concept slot j against zero with the other slots at zero (App. C). Sec. 5.4 provides the estimator for the reported effects.

Each causal alignment loss is based on these quantities grounded on dataset labels c_j rather than on predicted concept presence. Below, we outline each causal alignment term with their formulations in Eq. 3.

Necessity. Inactive concepts should have no effect on the model's output: Replacing an inactive concept slot j with a counterfactual should not change the output such that $\Delta^{\text{cf}}_j = 0$.

Sufficiency. Active concepts alone should produce the expert output: Starting from the all-zero output and adding each active concept's separate contribution should produce the expert output, $h_\phi(\mathbf{0}) + \sum_{j:\, c_j=1} \Delta^0_j = Y$.

Independence. Concept representations should contribute additively: Each concept should contribute on its own, and the per-concept contributions should sum to the total, $\Delta^0 = \sum_j \Delta^0_j$.

Presence (auxiliary). Since sufficiency constrains only the sum of concept representations, a subset of active concepts could carry all of the effect. To guard against that, we add an optional term that requires each active concept to carry at least an equal share $\beta = 1/\sum_j c_j$ of the total, $\|\Delta^0_j\| \geq \beta\|\Delta^0\|$ whenever $c_j = 1$.

$$\begin{aligned}
\mathcal{L}_{\text{nec}} &= \mathbb{E}\Big[\textstyle\sum_{j:\, c_j=0} \|\Delta^{\text{cf}}_j\|^2\Big], \quad & \mathcal{L}_{\text{suf}} &= \mathbb{E}\Big[\big\|h_\phi(\mathbf{0}) + \textstyle\sum_{j:\, c_j=1} \Delta^0_j - Y\big\|^2\Big], \\
\mathcal{L}_{\text{ind}} &= \mathbb{E}\Big[\big\|\Delta^0 - \textstyle\sum_j \Delta^0_j\big\|^2\Big], \quad & \mathcal{L}_{\text{pres}} &= \mathbb{E}\Big[\textstyle\sum_{j:\, c_j=1} \max\big(0,\, \beta\|\Delta^0\| - \|\Delta^0_j\|\big)^2\Big].
\end{aligned} \tag{3}$$

4.4 Training objective

We combine the four alignment losses into a single alignment term

$$\mathcal{L}_{\text{align}} = \lambda_{\text{nec}}\mathcal{L}_{\text{nec}} + \lambda_{\text{suf}}\mathcal{L}_{\text{suf}} + \lambda_{\text{ind}}\mathcal{L}_{\text{ind}} + \lambda_{\text{pres}}\mathcal{L}_{\text{pres}}, \tag{4}$$

and train the wrapper by minimizing

$$\mathcal{L}_{\text{wrap}}(\theta, \phi) = \mathcal{L}_{\text{task}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{align}}. \tag{5}$$

$\mathcal{L}_{\text{task}}$ is the supervised mean-squared error on $\hat{Y}$ (to the expert action for control, to the one-hot species label for CUB-200) and $\mathcal{L}_{\text{cls}}$ trains each concept slot's vector length to predict its concept label. Training proceeds in three phases (Algorithm 1): warmup on $\mathcal{L}_{\text{task}}$ and $\mathcal{L}_{\text{cls}}$, an intermission that fits the mediator model on the frozen encoder's concepts, and alignment on the full objective, with optional mediator model refits at fixed intervals. Loss weights, schedules, and architecture details are in Apps. E–G.

5 Empirical evaluation

We evaluate CCW-Net on three domains spanning control and perception: aircraft control, the CarRacing-v1 driving environment [Brockman et al., 2016], and image classification on CUB-200 [Wah et al., 2011].

5.1 Tasks

Aircraft control. We evaluate on extended trail formation [United States Air Force, 2024, 2025], in which a chase aircraft maintains position behind a lead aircraft executing dynamic maneuvers. The simulator is a high-fidelity aircraft model with six degrees of freedom [Heidlauf et al., 2018] where the lead aircraft follows a scripted trajectory and the chase aircraft is governed by a reinforcement learning policy trained by proximal policy optimization [Schulman et al., 2017] using the aircraft environment and codebase of So and Fan [2023]. The action space is two-dimensional (pitch and roll commands). Concept labels are $J = 4$ binary attributes (*Lead pursuit*, *Lag pursuit*, *Climb*, *Dive*) drawn from operational pilot vocabulary [United States Air Force, 2024, 2025]. Pursuit (*Lead pursuit*, *Lag pursuit*) and vertical (*Climb*, *Dive*) maneuver concept pairs are mutually exclusive, so exactly two concepts are active per sample. Full task details are in Appendix H.1.

CarRacing. We evaluate on the CarRacing-v1 environment [Brockman et al., 2016], in which a pixel-input agent drives a car around a track. The frozen backbone is a CNN policy trained end-to-end on the task. Its latents, actions, and concept labels come from a CW-Net test domain [Kenny et al., 2026]. The action space is continuous steering, throttle, and brake. Concept labels are $J = 3$ binary attributes (*Left*, *Right*, *Straight*). The concepts are mutually exclusive and exhaustive, so exactly one concept is active per sample and the presence threshold is $\beta = 1$. Full task details are in Appendix H.2.

CUB-200 classification. We evaluate on the CUB-200-2011 dataset of 200 bird species [Wah et al., 2011]. The frozen backbone is an Inception-v3 image classifier fine-tuned on CUB. Concept labels are $J = 112$ binary visual attributes (e.g., "has yellow belly", "has pointed beak") and the wrapper output is a 200-way classification. Full task details are in Appendix H.3.

5.2 Baselines

Same-architecture, no alignment. We compare CCW-Net against the same encoder and head trained without the alignment losses (only $\mathcal{L}_{\text{task}}$ and $\mathcal{L}_{\text{cls}}$ active). This isolates the contribution of the causal alignment objective from the contribution of the encoder architecture.

Concept Wrapper Network variants. On the aircraft and CarRacing tasks we compare against the Concept Wrapper Network (CW-Net) of Kenny et al. [2026], using the authors' architecture. CW-Net wraps a frozen backbone with a concept layer in the same spirit as CCW-Net, but implements a different architecture, represents each concept as a scalar, and trains without causal alignment. Appendix I adds CW-Net variants trained with alignment and, on CarRacing, with vector concepts.

Concept bottleneck model. For CUB-200 we compare against the concept bottleneck model of Koh et al. [2020], which trains a model to predict concepts and to derive the task output through them.

Disentanglement-based concept learning. To support the secondary claim that disentanglement does not imply causal alignment, we compare against β-TCVAE [Chen et al., 2018] and iVAE [Khemakhem et al., 2020] on the aircraft and CarRacing tasks, reporting DCI [Eastwood and Williams, 2018] alongside the concept effect fractions (Appendix I).

5.3 Ablations

We run five ablation types on the control domains: (i) **loss combinations** where subsets of the four alignment losses are disabled to test each one's contribution, (ii) **concept representation dimension** with $D = 16$ default, $D = 1$ scalar, and $D = 8$, (iii) **data quantity** for aircraft ranging from 5K to 5M samples, (iv) **label quality** as fraction of data labeled with concepts and injected label noise, and (v) **concept ontology** where we remove concepts, merge concepts, and add a placebo concept. The full ablation matrix is in Appendix I. The CUB-200 models train without the alignment losses and compare architectures (Sec. 5.5.3).

5.4 Evaluation methodology

We report task performance (mean-squared error for action regression), concept accuracy against data labels, DCI [Eastwood and Williams, 2018], and the causal effect each concept has on the model's output. All values are means over ten seeds with two-sided 95% Student-t intervals. The primary metrics in this study are causal effects.

Causal attribution. To test whether a model's concept representations carry only their own effects, we estimate interventional indirect effects [Vansteelandt and Daniel, 2017, Loh et al., 2022] from held-out observational data, using no mediator model, no zero reference $\mathbf{0}$, and none of the training losses. The approach applies to any model whose output is a function of its concept representations. We use it on CCW-Net, CW-Net, and the disentanglement baselines.

Each comparison takes two sets of test inputs, $\mathcal{A}$ and $\mathcal{B}$, whose concept labels differ where an active concept in $\mathcal{A}$ is replaced in $\mathcal{B}$ by a mutually exclusive alternative with all other labels unchanged. For a test input i, we draw matches from $\mathcal{A}$ and $\mathcal{B}$ by Gaussian-kernel matching on z-scored features z (raw inputs or for image inputs, backbone features $f(X)$). A test input $i' \neq i$ is sampled with probability proportional to $\exp(-d_{ii'}^2/2\tau^2)$ among those within a radius of i, where $d_{ii'}$ is the root-mean-square distance between z_i and $z_{i'}$ and τ is the kernel bandwidth. If none are within the radius, we take the nearest neighbor. The matches' concepts $g_\theta(f(X_{i'}))$ fill the concept slots. The two slots whose labels differ form one group, so every hybrid has a label combination that occurs in the data, and every other slot is a group of its own. We then build hybrid concept vectors taking each group of concept mediator representation slots from the $\mathcal{A}$ or the $\mathcal{B}$ match, and pass them through the trained head h_ϕ. Group G's effect is the change in output when it switches from its $\mathcal{A}$ match to its $\mathcal{B}$ match,

$$\mathrm{IE}_G = \mathbb{E}_{i,\pi}\Big[h_\phi\big(M^{\mathcal{B}}_{\pi<G},\, M^{\mathcal{B}}_{G},\, M^{\mathcal{A}}_{\pi>G}\big) - h_\phi\big(M^{\mathcal{B}}_{\pi<G},\, M^{\mathcal{A}}_{G},\, M^{\mathcal{A}}_{\pi>G}\big)\Big], \tag{6}$$

where $M^{\mathcal{A}}_G$ ($M^{\mathcal{B}}_G$) are group G's slots from an $\mathcal{A}$ ($\mathcal{B}$) match, drawn independently for each group, and likewise for the groups $\pi < G$ ($\pi > G$) before (after) G in an ordering π. Averaging over all orderings makes IE_G group G's Shapley value for the switch from $\mathcal{A}$ to $\mathcal{B}$, and i runs over the inputs of both sets. Estimator details are in App. F.

For each label combination (rows in Tables 1 and 2), slot j's effect $\|\mathrm{IE}_j\|$ is the norm of its own group's effect in the comparisons that keep j's label fixed, i.e., how far slot j moves the output when another concept changes. The fractions of the effect that slot j, the active concepts, and the inactive concepts carry are

$$\rho_j = \frac{\|\mathrm{IE}_j\|}{\sum_k \|\mathrm{IE}_k\|}, \qquad \rho_{\text{active}} = \sum_{j:\, c_j=1} \rho_j, \qquad \rho_{\text{inactive}} = \sum_{j:\, c_j=0} \rho_j = 1 - \rho_{\text{active}}. \tag{7}$$

Inactive concepts should carry no effect, so a faithful model has ρ_{inactive} near zero.

Table 1: Aircraft control: fraction of the effect each concept carries (Eq. 7), per concept label combination, estimated as in Sec. 5.4. Active concepts are in bold. All Active and All Inactive are ρ_{active} and ρ_{inactive}. Each block's header gives the model's test action MSE (tMSE), concept accuracy on the Lead/Lag and Climb/Dive concept sets (%), and DCI disentanglement / completeness / informativeness. Means over ten seeds with two-sided 95% Student-t intervals. CCW-Net shows superior causal effect routing through active concepts compared to prior art.

	Concept Effect (%)					
Label combination	Lead	Lag	Climb	Dive	All Active ↑	All Inactive ↓
CCW-Net (Ours)						
tMSE 0.125 ±0.001; concept acc. 95.7 ±0.1 / 99.3 ±0.0; DCI 0.999 ±0.001 / 0.719 ±0.041 / 0.975 ±0.001						
Lead+Climb	**62.8** ±16.1	4.6 ±1.9	**32.0** ±16.1	0.6 ±0.4	94.8 ±2.1	5.2 ±2.1
Lead+Dive	**69.2** ±15.7	5.4 ±2.9	1.0 ±0.5	**24.4** ±16.0	93.6 ±3.3	6.4 ±3.3
Lag+Climb	4.3 ±2.3	**60.5** ±13.2	**34.5** ±13.5	0.7 ±0.4	95.0 ±2.5	5.0 ±2.5
Lag+Dive	4.9 ±2.4	**67.8** ±13.0	1.1 ±0.6	**26.2** ±13.3	94.0 ±2.6	6.0 ±2.6
CCW-Net, no causal alignment losses						
tMSE 0.124 ±0.001; concept acc. 95.7 ±0.1 / 99.3 ±0.0; DCI 0.999 ±0.000 / 0.746 ±0.037 / 0.975 ±0.001						
Lead+Climb	**63.7** ±21.0	9.5 ±6.0	**25.0** ±16.3	1.8 ±2.0	88.7 ±7.2	11.3 ±7.2
Lead+Dive	**64.8** ±20.0	9.6 ±5.6	4.1 ±3.9	**21.5** ±14.0	86.3 ±6.8	13.7 ±6.8
Lag+Climb	8.0 ±2.8	**62.6** ±16.9	**27.3** ±15.6	2.0 ±2.0	89.9 ±3.7	10.1 ±3.7
Lag+Dive	8.5 ±4.3	**63.4** ±17.9	5.2 ±6.3	**22.9** ±14.6	86.3 ±6.3	13.7 ±6.3
CW-Net [Kenny et al., 2026]						
tMSE 0.122 ±0.002; concept acc. 91.2 ±0.2 / 98.2 ±0.1; DCI 0.998 ±0.001 / 0.719 ±0.064 / 0.947 ±0.001						
Lead+Climb	**30.5** ±11.8	41.0 ±11.0	**10.6** ±4.5	17.8 ±10.6	41.2 ±12.7	58.8 ±12.7
Lead+Dive	**25.2** ±11.2	31.4 ±8.6	17.3 ±6.6	**26.0** ±9.8	51.3 ±6.3	48.7 ±6.3
Lag+Climb	31.5 ±13.2	**38.8** ±13.2	**11.5** ±4.7	18.2 ±8.4	50.3 ±13.0	49.7 ±13.0
Lag+Dive	26.1 ±13.1	**28.2** ±9.2	18.3 ±6.5	**27.4** ±8.5	55.6 ±14.1	44.4 ±14.1

On CUB-200, we report a separate per-input evaluation. Its 112 attribute labels are assigned per species and have no mutually exclusive alternatives within the dataset, so the comparisons above, which hold all other labels fixed, cannot be formed with enough test inputs. Instead, each concept in turn is set to absent (the zero slot for CCW-Net, concept probability 0 for CBM), ρ_j is its share of the summed output changes at each test input, and we average over the test split. This uses the zero reference, but no CUB-200 model is trained with the alignment losses, so it reuses no optimized terms.

5.5 Results

5.5.1 Aircraft control

Table 1 reports, for each concept label combination, the fraction of the effect each concept representation mediator carries. CCW-Net (concept representation dimension $D = 16$) routes $94.5 \pm 2.2\%$ of the effect through the active concepts, versus $88.1 \pm 4.3\%$ trained without the causal alignment losses maintaining comparable task error and concept accuracy. CW-Net predicts its concepts nearly as accurately (91.2% and 98.2% on the two mutually exclusive concept sets), yet its active concepts carry only $50.0 \pm 6.2\%$ of the effect. This is the fraction expected if the effect were spread indiscriminately across the four concepts. This suggests accurate concept prediction alone does not make the model's outcomes depend on them. We find DCI disentanglement is near its ceiling of 1 for every model in Table 1 suggesting that disentanglement metrics may be insufficient to imply causal relationships between representation and model outcomes. The disentanglement baseline iVAE reaches DCI disentanglement, completeness and informativeness of 1.000, 0.891 and 1.000, yet its active concepts carry only $47.6 \pm 2.4\%$ of the effect (App. I.1).

5.5.2 CarRacing

Table 2 reports the same for CarRacing, where one of the three concepts is active in each label combination. CCW-Net (concept representation dimension $D = 1$) routes $89.8 \pm 2.5\%$ of the effect through the active concept, versus $87.5 \pm 4.9\%$ trained without the causal alignment losses at the

Table 2: CarRacing: fraction of the effect each concept carries (Eq. 7), per concept label combination, estimated as in Sec. 5.4. Active concepts are in bold. All Active and All Inactive are ρ_{active} and ρ_{inactive}. Each block's header gives the model's test action MSE (tMSE), concept accuracy on the Straight/Left/Right concept set (%), and DCI disentanglement / completeness / informativeness. Means over ten seeds with two-sided 95% Student-t intervals. CCW-Net shows superior causal effect routing through active concepts compared to prior art.

	Concept Effect (%)				
Label combination	Straight	Left	Right	All Active ↑	All Inactive ↓
CCW-Net (Ours)					
tMSE 0.0419 ±0.0005; concept acc. 92.6 ±0.4; DCI 0.863 ±0.023 / 0.863 ±0.023 / 0.943 ±0.005					
Straight	**98.1** ±1.1	1.3 ±0.9	0.5 ±0.4	98.1 ±1.1	1.9 ±1.1
Left	25.2 ±7.0	**74.5** ±7.0	0.3 ±0.2	74.5 ±7.0	25.5 ±7.0
Right	23.5 ±5.8	0.7 ±0.4	**75.8** ±5.8	75.8 ±5.8	24.2 ±5.8
CCW-Net, no causal alignment losses					
tMSE 0.0419 ±0.0006; concept acc. 92.4 ±0.4; DCI 0.850 ±0.021 / 0.851 ±0.021 / 0.944 ±0.005					
Straight	**95.4** ±3.3	3.5 ±2.5	1.2 ±0.9	95.4 ±3.3	4.6 ±3.3
Left	26.2 ±9.2	**73.2** ±9.4	0.6 ±0.3	73.2 ±9.4	26.8 ±9.4
Right	24.5 ±7.5	1.7 ±1.0	**73.9** ±7.9	73.9 ±7.9	26.1 ±7.9
CW-Net [Kenny et al., 2026]					
tMSE 0.0015 ±0.0007; concept acc. 90.0 ±1.3; DCI 0.078 ±0.063 / 0.626 ±0.082 / 0.928 ±0.008					
Straight	**33.2** ±1.7	37.4 ±7.4	29.4 ±6.8	33.2 ±1.7	66.8 ±1.7
Left	33.7 ±11.0	**32.9** ±1.4	33.4 ±10.4	32.9 ±1.4	67.1 ±1.4
Right	29.7 ±7.0	37.3 ±6.5	**33.1** ±2.1	33.1 ±2.1	66.9 ±2.1

Table 3: CUB-200: fraction of the effect the active concepts carry, per input (Sec. 5.4), averaged over the test split. CBM is the concept bottleneck model of Koh et al. [2020]. No CUB-200 model is trained with the causal alignment losses. Concept accuracy is over the 112 attributes. Means over ten seeds with two-sided 95% Student-t intervals.

	Concept	Concept Effect (%)		DCI		
Model	acc. (%)	All Active ↑	All Inactive ↓	Disentanglement	Completeness	Informativeness
CCW-Net (Ours)	99.6 ±0.0	91.3 ±0.3	8.7 ±0.3	0.895 ±0.007	0.936 ±0.007	0.993 ±0.001
CBM	97.7 ±0.1	41.6 ±0.6	58.4 ±0.6	0.732 ±0.011	0.789 ±0.011	0.985 ±0.001

same task error and comparable concept accuracy, so on CarRacing the architecture accounts for most of the causal routing. While CW-Net reaches lower task error than CCW-Net, both policies were able to complete the task upon visual inspection. While both approaches predict their concepts nearly as accurately ($92.6 \pm 0.4\%$ and $92.4 \pm 0.4\%$ vs $90.0 \pm 1.3\%$) CW-Net's active concept carries only $33.1 \pm 1.6\%$ of the effect, the fraction expected if the effect were spread indiscriminately across the three concepts.

5.5.3 CUB-200

Table 3 reports CUB-200. Neither CCW-Net nor CBM is trained with the causal alignment losses, so this is a comparison that isolates the architecture. CCW-Net (concept representation dimension $D = 1$) routes $91.3 \pm 0.3\%$ of the effect through active concepts, versus $41.6 \pm 0.6\%$ for CBM with both predicting their concepts accurately ($99.6 \pm 0.0\%$ and $97.7 \pm 0.1\%$). More than half of CBM's effect ($58.4 \pm 0.6\%$) is attributed to concepts labeled absent. Similar to the above studies, accurate concept prediction alone does not make the model's outcomes depend on the active concepts. In CCW-Net, a zeroed concept vector carries a zero key and zero value in the attention head, so setting a concept to absent removes its contribution to the outcome $\hat{Y}$ (Prop. 1).

5.5.4 Robustness and ablation

We performed ablation studies on the control domains: aircraft control and CarRacing. Full results are in Tables 8 and 9 (App. I).

Label efficiency. Across both control domains, CCW-Net retains most of its causal faithfulness when concept supervision is increasingly sparse. For example, when only 10% of data is labeled with concepts, ρ_{active} decreases by only 4.0 points on aircraft control (97.3% to 93.3%) and interestingly increases by 2.6 on CarRacing (75.3% to 77.9%). Both are within noise across seeds. We believe this is consistent with CCW-Net's alignment losses and architecture providing strong enough inductive bias that dense labeling becomes less critical. These results are encouraging as they suggest that less labeling burden is required to retain comparable causal alignment results.

Label noise. We find that label noise has the opposite effect. 10% noise drops ρ_{active} from 97.3% to 71.6% on aircraft control and from 75.3% to 61.8% on CarRacing while concept accuracy remains nearly unchanged (96.2% vs. 96.4% and 92.3% vs. 93.4%). This is an example where concept accuracy and architecture alone do not reflect causal faithfulness motivating the need for evaluation approaches such as ours to verify such claims.

Ontology misspecification. CCW-Net degrades gracefully when the ontology isn't fully specified. Removing one concept from the aircraft ontology lowers ρ_{active} to 79.9–94.6%, and merging two concepts leaves it at 95.6–97.4%. On CarRacing, removing one concept gives 65.8–82.1% and merging two gives 56.9–95.7%, against 75.3% for the full ontology.

Loss ablation. On aircraft control, removing the sufficiency or independence loss terms lowers ρ_{active} from 97.3% to 94.4% and 94.8%, while removing the presence term does not (98.0%). On CarRacing, where one concept is active per input, necessity alone reaches 77.2%, comparable to 75.3% with all four losses, and presence alone reaches 71.9% against 63.5% without alignment. This suggests different domains benefit from the loss terms differently.

Vector versus scalar concepts. We find concept representation dimension $D > 1$ allows the alignment losses flexibility to improve causal alignment. At $D = 16$, alignment increases ρ_{active} by 8.1 points on aircraft control and 11.8 points on CarRacing, while at $D = 1$ the change is -0.9 and 2.4 points.

6 Conclusion

Concept-based explanations are useful to the extent that they causally drive the model's outputs. We presented CCW-Net, the first formulation of multi-mediator interventional mediation analysis as a training objective for deep networks. We also introduced an evaluation method that quantifies the fraction of a model's output effect concept explanations carry. CCW-Net wraps a frozen backbone model with a concept encoder and attention head, trained to satisfy three causal alignment criteria (necessity, sufficiency, and independence) with counterfactual concept representations drawn from a mediator model. CCW-Net's causal claims concern the effect of concept activations and representations on model outcomes where effects in the data-generation process are left to future work.

Across aircraft control, CarRacing-v1, and CUB-200 image classification, CCW-Net routes more of its output's effect through active concepts than concept bottleneck based approaches. The contribution combines two approaches: concept representations feeding an attention head in which absent concepts contribute zero value, isolating each concept's contribution to the output, and causal alignment losses that strengthen this effect and improve consistency across seeds. On our aircraft control domain, CCW-Net routes $94.5 \pm 2.2\%$ of the effect through active concepts against $50.0 \pm 6.2\%$ for prior art. On CUB-200, the architecture alone routes 91.3% against 41.6% for CBM at comparable concept accuracy, suggesting that concept accuracy is not an indicator of causal faithfulness. iVAE achieves excellent DCI disentanglement metrics on aircraft control, yet its active concepts carry only 47.6% of the effect, suggesting disentanglement and causal alignment are different properties.

CCW-Net moves causal faithfulness from a property that is claimed by inspection to one that can be designed for, trained against, and measured by. Concept-based explanations become not merely interpretable but quantifiably causal thereby building a road toward verifiably trustworthy models in safety- and task-critical environments.

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A Variable dictionary

Table 4: Variable dictionary for CCW-Net.

Symbol	Definition
Inputs and outputs	
X	Input
Y	Expert output
$\hat{Y}$	Wrapper output, $h_\phi(M)$
$c_j \in \{0, 1\}$	Concept label ($1 =$ active)
Architecture	
f	Frozen backbone
$Z = f(X)$	Backbone latent
g_θ	Concept encoder
h_ϕ	Attention head
$M = (M_1, \ldots, M_J)$	Concept representation
$M_j \in \mathbb{R}^D$	Concept j's representation (slot)
J	Number of concepts
D	Representation dimension
Mediator model (Sec. 4.2)	
$\mathcal{F}_\psi$	Per-concept invertible flow
$\tilde{M}_j$	Counterfactual draw, $\tilde{M}_j \sim \mathcal{F}_\psi(\cdot \mid 1 - c_j, X)$
$\mathcal{L}_{\text{flow}}$	Flow negative log-likelihood
Mediation analysis (Sec. 3)	
TE, DE, IE	Total, direct, indirect effect
IE_j	Interventional indirect effect of mediator j
IE_μ	Residual mediator interaction
Concept edits and changes (Sec. 4.3)	
$\mathbf{0}$	All-zero concept representation
$\mathbf{0}_{j \leftarrow M_j}$	$\mathbf{0}$ with slot j set to M_j
$M_{j \leftarrow \tilde{M}_j}$	M with slot j set to $\tilde{M}_j$
Δ^0	Change from zero, $\hat{Y} - h_\phi(\mathbf{0})$
Δ^0_j	Slot j's change from zero, $h_\phi(\mathbf{0}_{j \leftarrow M_j}) - h_\phi(\mathbf{0})$
Δ^{cf}_j	Slot j's counterfactual change, $h_\phi(M_{j \leftarrow \tilde{M}_j}) - \hat{Y}$
Losses and weights	
$\mathcal{L}_{\text{task}}$	Task loss
$\mathcal{L}_{\text{cls}}$	Concept classification loss
$\mathcal{L}_{\text{nec}}, \mathcal{L}_{\text{suf}}, \mathcal{L}_{\text{ind}}$	Necessity, sufficiency, independence losses
$\mathcal{L}_{\text{pres}}$	Presence loss (auxiliary)
$\mathcal{L}_{\text{align}}$	Weighted sum of the alignment losses
$\mathcal{L}_{\text{wrap}}$	Total wrapper loss
$\lambda_{\text{cls}}, \lambda_{\text{nec}}, \ldots, \lambda_{\text{pres}}$	Loss weights
β	Presence threshold, $1/\sum_j c_j$
Evaluation (Sec. 5.4)	
$\mathcal{A}, \mathcal{B}$	Test-input sets of one comparison
z_i	Matching features of input i
$d_{ii'}$	Distance between z_i and $z_{i'}$
τ	Kernel bandwidth
G, π	Slot group; group ordering
$M^{\mathcal{A}}_G, M^{\mathcal{B}}_G$	Group G's slots from an $\mathcal{A}$ or $\mathcal{B}$ match
IE_G	Group G's effect
ρ_j	Fraction of the effect slot j carries
$\rho_{\text{active}}, \rho_{\text{inactive}}$	Fraction active, inactive concepts carry
Evaluation estimator (App. F)	
ℓ, n_ℓ	Concept label combination, its test count
c, $\mathcal{C}(\ell)$	Comparison, comparisons that include label combination ℓ
G^{sw}_c	Swapped pair of comparison c

Symbol	Definition
$\mathcal{S}$	One of the sets $\mathcal{A}, \mathcal{B}$
i'	Match of test input i
K	Number of matching features
ϵ	Matching radius
$\mathrm{IE}_G^{(c)}$	Group G's effect in comparison c
$\mathcal{C}_j(\ell)$	Comparisons in $\mathcal{C}(\ell)$ with $j \notin G_c^{\mathrm{sw}}$
$G_c(j)$	Concept slot j's group in comparison c
$\|\mathrm{IE}_j\|^{(\ell)}, \rho_j^{(\ell)}$	Concept slot j's effect, fraction for label combination ℓ
Λ	Scored combinations with an active concept
$\bar{\rho}_{\text{active}}, \bar{\rho}_{\text{inactive}}$	Test-count-weighted means over Λ

B Extended related work

In Section 2 we focused on the most directly relevant prior art for CCW-Net's three contributions (a mediation-analysis training objective, an architecture for concept-level mediation analysis, and an evaluation method for concept-based causality claims). Here we position CCW-Net among causal concept bottleneck models and causal abstraction methods, and discuss adjacent research operating at different levels (input, environment-state, policy, trajectory) or in adjacent approaches.

B.1 Causal concept bottleneck models and causal abstraction

Concept bottleneck models differ in how concepts reach and produce the model's output. In flat bottlenecks, every concept is a direct parent of the output. Laguna et al. [2024] make pretrained black boxes intervenable, fine-tuning them such that concept interventions improve their predictions. Ismail et al. [2024] insert a concept bottleneck into generative models to steer generation toward human-interpretable concepts. Dominici et al. [2025b] extend concept bottleneck models to predict how concepts would need to change to produce an alternative prediction. These works use concept interventions to improve or steer predictions. CCW-Net instead trains and measures how much of a model's output is attributed to each concept. Concept graph models [De Felice et al., 2025, Dominici et al., 2025a] let some concepts be parents of others rather than of the output, allowing for a deeper, more intricate concept graph. Driving and measuring the causal relationship of each node in such a graph is a natural extension of our approach.

In the causal abstraction literature, interchange intervention training [Geiger et al., 2022] trains a network so that interventions on aligned high-level variables produce the counterfactual outputs of a manually-specified causal model. Causal proxy models [Wu et al., 2023] train a proxy of a text classifier on counterfactual examples so that concept interventions simulate the classifier's counterfactual behavior. CCW-Net needs neither a hand-specified causal model nor counterfactual input pairs. Its losses follow interventional mediation analysis [Vansteelandt and Daniel, 2017], which also requires no causal ordering among the mediators. For measurement of causal effects, the causal concept effect (CaCE) of Goyal et al. [2019] estimates the change in expected output when a concept is toggled, using a learned generative model of counterfactual inputs and averaging over all samples. CCW-Net trains for the property and measures it separately for active and inactive concepts.

B.2 Causal explanation in sequential decision-making

Several lines of work explain sequential decision-making policy behavior using causal frameworks at the environment-state, trajectory, or action level. Madumal et al. [2020] introduce action-influence models that define a structural causal model over environment variables and agent actions and use them to produce contrastive explanations of RL policies. Kekić et al. [2026] learn nonlinear causal reductions that map a high-dimensional RL system to a lower-dimensional SCM whose interventional behavior matches the original system. Nashed et al. [2025] build a unified SCM framework for explaining the actions of MDP agents, and Gyevnar et al. [2024] extend causal explanations to multi-agent settings. Bica et al. [2021] explain expert sequential decision-making policies by learning rewards over what-if outcomes. Schulte and Poupart [2025] examine when reinforcement learning benefits from causal reasoning. Verma and Srivastava [2024] learn causally accurate models for autonomous assessment of black-box agents. These methods operate at the environment or policy level

and do not impose causal structure on a learned concept representation. CCW-Net is complementary: it uses concept labels to shape the model's own concept-to-output effects, without requiring an explicit environment SCM.

B.3 Causal methods in imitation and reinforcement learning training

Causal frameworks have also been incorporated into imitation learning and reinforcement learning training, distinct from post-hoc explanation. Ruan et al. [2023] introduce causal imitation learning via inverse reinforcement learning to recover policies that respect causal structure in expert demonstrations. Li et al. [2025] address confounding-robust deep RL via a causal approach to off-policy learning. Herlau and Larsen [2022] learn causal variables and a policy by optimizing a mediation-analysis objective in which the treatment is the choice of policy, the mediator is a binary environment event, and the outcome is the return. These works apply causal reasoning at the policy or imitation level. CCW-Net applies causal reasoning at the concept level inside the policy, training the concept encoder to satisfy interventional necessity, sufficiency, and independence criteria.

B.4 Human-AI interaction, trust, and verification

Several lines of work motivate interpretable models in safety-critical settings from human-factors and verification perspectives. Sanneman and Shah [2022], Sanneman et al. [2024] characterize the role of explanations in supporting human situational awareness and the understanding-workload tradeoff in human-centered XAI. Sreedharan et al. [2022] provide a planning perspective on explainable human-AI interaction. Sobrín-Hidalgo et al. [2025] survey explanation generation for autonomous robots. Lyons et al. [2024], Highland et al. [2023] report empirical studies of trust and agent appropriateness in live-flight human-autonomy teaming. Omeiza et al. [2022] survey explanation methods specific to autonomous driving. Verma and Shah [2025], Mahmud et al. [2024] consider verification and interpretability analysis of decision-making systems. CCW-Net sits within this broader effort: it produces explanations grounded in human-meaningful concepts and aligned with causal effects, supporting the verification, audit, and trust workflows these works study.

C Connection to interventional mediation analysis

In this section, we connect CCW-Net's method and alignment losses (Sec. 4.3) to interventional mediation analysis [Vansteelandt and Daniel, 2017, Loh et al., 2022], and state the attention head property that makes $M_j = 0$ a valid reference (App. C.1).

Interventional mediation analysis. Mediation analysis asks how much of a treatment's effect on an outcome passes through each of several intermediate variables (i.e., the mediators). Following the notation of Vansteelandt and Daniel [2017], we write Y for the outcome and $M = (M_1, \ldots, M_J)$ for the mediators. $M(a)$ denotes the mediator values under treatment a, and $Y(a, m)$ denotes the outcome under treatment a with the mediators set to m. Changing the treatment from a to a^* has the total effect

$$\mathrm{TE} = \mathbb{E}\big[Y(a^*, M(a^*)) - Y(a, M(a))\big]. \tag{8}$$

Vansteelandt and Daniel [2017] measure the part of this effect that passes through a single mediator M_j with the interventional indirect effect

$$\mathrm{IE}_j = \mathbb{E}\big[Y(a^*, M_j(a^*), M_{-j}(a)) - Y(a^*, M_j(a), M_{-j}(a))\big]. \tag{9}$$

IE_j sets only M_j to its value under the new treatment and keeps every other mediator at its value under the original treatment. The direct effect $\mathrm{DE} = \mathbb{E}[Y(a^*, M(a)) - Y(a, M(a))]$ is the part of the effect that bypasses the mediators. Vansteelandt and Daniel [2017] and Loh et al. [2022] add a residual term IE_μ for the interactions among mediators. With it, the effects add up to the total without assuming any causal ordering or independence among the mediators,

$$\mathrm{TE} = \mathrm{DE} + \sum_{j=1}^{J} \mathrm{IE}_j + \mathrm{IE}_\mu. \tag{10}$$

Mapping to CCW-Net. In CCW-Net, the concept representations M_j are the mediators and the head output $\hat{Y}$ is the outcome. As the concept representations serve as a bottleneck, the input reaches $\hat{Y}$ only through them with no direct pathway from input to output that bypasses the mediators, so $\mathrm{DE} = 0$ (Sec. 4.1). Flipping concept j's label plays the role of the treatment change, and the flow turns this flip into the counterfactual draw $\tilde{M}_j$ (Eq. 1). In the notation above, the treatment value is concept j's label, $a = c_j$, and $\tilde{M}_j$ plays the role of $M_j(a^*)$ with $a^* = 1 - c_j$. With this mapping, two of the changes in Sec. 4.3 correspond to IE_j.

- Δ_j^{cf} is the per-sample analog of IE_j. Like IE_j, it replaces M_j with its counterfactual value and keeps the other concept representations at their observed values.
- Δ_j^0 is the analog of IE_j measured from zero. It switches M_j on with every other concept representation at zero instead of at its observed value. Zero is a valid reference because $h_\phi(\mathbf{0})$ is a constant (Prop. 1).

What each loss enforces.

- *Necessity.* $\mathcal{L}_{\mathrm{nec}}$ drives Δ_j^{cf} toward zero for concepts labeled inactive, so an inactive concept has no per-sample indirect effect on the output.
- *Sufficiency.* $\mathcal{L}_{\mathrm{suf}}$ anchors $h_\phi(\mathbf{0}) + \sum_{j:\, c_j=1} \Delta_j^0$ to the expert output Y. Since $\mathrm{DE} = 0$, the active concepts' contributions measured from zero should reproduce the expert output.
- *Independence.* $\mathcal{L}_{\mathrm{ind}}$ drives $\Delta^0 - \sum_j \Delta_j^0$ toward zero. This residual plays the role of IE_μ in Eq. (10) for changes measured from zero, so driving it to zero makes the concept contributions add up.

$\mathcal{L}_{\mathrm{pres}}$ is not part of the decomposition. It keeps a few active concepts from carrying the whole effect. Without it, sufficiency could be met overall while the individual contributions say little.

C.1 Zero value contribution of a zeroed concept

Zero can only serve as a reference if a zeroed concept representation contributes nothing to the output. Here we show that the attention head has this property, and we describe what zeroing a concept does to the other concepts' attention weights.

Setup. The wrapper output $\hat{Y}$ is a vector indexed by r. The head is a single-head attention layer with one learned query q_r per output. W_Q, W_K, W_V, and W_O are the head's query, key, value, and output projections, all of which are bias-free, and w_r and b_r are the weight and bias for output r. We use j, k, and l to index the J concept representations, where j is the one intervened on, k is any other, and l is a summation index. For output r, the head computes logits $s_{rj} = (W_Q q_r)^\top W_K M_j / \sqrt{D}$, weights $\alpha_{rj} = e^{s_{rj}} / \sum_l e^{s_{rl}}$, context $c_r = \sum_l \alpha_{rl} W_O W_V M_l$, and output $\hat{Y}_r = \tanh(w_r^\top c_r + b_r)$.

Definition 1 (Zero value contribution)**.** *Concept j's value contribution to output r is $\kappa_{rj}(M) = \alpha_{rj}(M)\, W_O W_V M_j$. The head has zero value contribution if $\kappa_{rj}(M) = 0$ for every concept j, every output r, and every M with $M_j = 0$.*

Proposition 1. *(i) The head has zero value contribution, and its output at $M = \mathbf{0}$ is the constant $\hat{Y}_r = \tanh(b_r)$. (ii) A concept with $M_j = 0$ keeps softmax weight $\alpha_{rj} = 1/Z_r > 0$ with $Z_r = \sum_{l \neq j} e^{s_{rl}} + 1$, so zeroing a concept is not the same as deleting it. (iii) Any single-concept intervention $M_j \to M_j'$ rescales the other concepts' weights by one common factor per output, $\alpha'_{rk} = \gamma_r \alpha_{rk}$ for all $k \neq j$, with*

$$\gamma_r = \frac{\sum_{l \neq j} e^{s_{rl}} + e^{s_{rj}}}{\sum_{l \neq j} e^{s_{rl}} + e^{s'_{rj}}}.$$

For the zero intervention $M_j' = 0$, the denominator equals the Z_r of (ii), and the effect on the context is exactly $c_r - c_r' = \kappa_{rj} + (1 - \gamma_r)(c_r - \kappa_{rj})$.

Proof. (i) The projections are linear, so $M_j = 0$ gives $W_V M_j = 0$ and hence $\kappa_{rj} = \alpha_{rj} W_O W_V M_j = 0$ for every r, whatever the value of α_{rj}. At $M = \mathbf{0}$, every logit is 0, the

weights are uniform, and $c_r = 0$, so $\hat{Y}_r = \tanh(b_r)$. (ii) A zero key gives a zero logit rather than a zero weight, since $e^0 = 1$. Deleting the concept would instead give the denominator $Z_r - 1$. (iii) Each logit depends only on its own concept, so the intervention changes only s_{rj}. For $k \neq j$, the ratio α'_{rk}/α_{rk} is the ratio of the two denominators, which does not depend on k. The zeroed concept's own term vanishes by (i), so $c'_r = \gamma_r(c_r - \kappa_{rj})$, which gives the identity. □

In words, zeroing a concept removes its own value term exactly. Its attention weight stays positive, and the other concepts' weights all scale by the same factor γ_r.

Relation to ignorability. Prop. 1 complements the ignorability condition in App. D. Ignorability (Prop. 2) handles confounding between the mediators and the outcome, $\hat{Y}(m) \perp M \mid X$. Prop. 1 gives every intervention a zero reference that itself contributes no value to the output.

D Causal assumptions

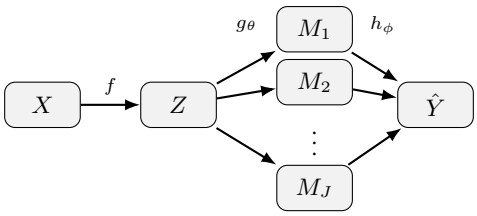

Figure 2: The CCW-Net wrapper SCM. Every path from X to $\hat{Y}$ passes through the concept representation M, and there is no direct $Z \to \hat{Y}$ edge. This gives mediator-outcome ignorability within the SCM (Prop. 2).

D.1 Identification assumptions for interventional effects

Both training and evaluation estimate interventional effects in the wrapper SCM

$$X \to Z \to M \to \hat{Y},$$

where $Z = f(X)$, $M = g_\theta(Z)$, and $\hat{Y} = h_\phi(M)$ (Fig. 2). Following Vansteelandt and Daniel [2017] and Loh et al. [2022], these effects are identified under the following standard conditions.

- **Ignorability.** There is no unmeasured confounding of the mediator-outcome relation given X, that is, $\hat{Y}(m) \perp M \mid X$ in the wrapper SCM. Prop. 2 shows that this holds by construction.
- **Positivity.** For every x in the support of X, the conditional distribution $p(M_j \mid X = x)$ has support around the observed concept representations. This condition concerns the counterfactual draws that the mediator model supplies to the necessity loss during training. The zero reference used by the sufficiency and independence losses is instead defined by the architecture (Prop. 1), not a property that occurs in the data.
- **Overlap.** In evaluation (Sec. 5.4), the mediator model is replaced with real data test inputs. For each test input i, both sets $\mathcal{A}$ and $\mathcal{B}$ should contain inputs within a radius of i, so that the matched concept representations come from comparable state configurations. When no input is within the radius, the estimator uses the nearest neighbor.
- **Consistency.** For the realized concept representation $M = m$, the observed $\hat{Y}$ equals $\hat{Y}(m)$ under the head h_ϕ.

Under these conditions, the effects decompose as in Eq. (10), with IE_μ collecting the interactions that cannot be attributed to any single concept [Loh et al., 2022].

D.2 Model-level ignorability in the wrapper SCM

The ignorability condition above holds by construction in the wrapper SCM. The following proposition shows this holds even when the wrapper has stochastic exogenous inputs such as in stochastic policies or with dropout.

Proposition 2 (Model-level mediator-outcome ignorability)**.** *Consider the wrapper SCM with structural equations*

$$Z = f(X, U_f), \quad M = g_\theta(Z, U_g), \quad \hat{Y} = h_\phi(M, U_h),$$

where U_f, U_g, U_h are exogenous noise variables. Suppose:

1. *$U_h \perp (X, U_f, U_g)$, so the head's noise is independent of everything upstream.*
2. *All randomness used for counterfactual sampling from the mediator model $\mathcal{F}_\psi$ depends only on (X, U_s) with $U_s \perp U_h$.*

Then for every concept representation value m in the support of $M \mid X$,

$$\hat{Y}(m) \perp M \mid X.$$

Hence the mediator-outcome ignorability required to identify IE_j*,* TE*, and* IE_μ *holds within the wrapper SCM.*

Proof. For a fixed m, the potential outcome $\hat{Y}(m) = h_\phi(m, U_h)$ is a function of U_h alone. M is a function of (X, U_f, U_g), and U_h is independent of (X, U_f, U_g) by assumption 1. Hence $\hat{Y}(m)$ is independent of (X, M), and in particular $\hat{Y}(m) \perp M \mid X$. A counterfactual draw is a function of (X, U_s), and $U_s \perp U_h$ by assumption 2, so the same argument applies with the draw in place of M. □

E Architecture details

E.1 Concept encoder g_θ

The concept encoder is a three-layer multi-layer perceptron with ReLU activations. It maps the backbone latent $Z = f(X)$ to J concept representations,

$$g_\theta(Z) = \mathrm{MLP}(Z) \in \mathbb{R}^{J \times D}. \tag{11}$$

The hidden width is 64 for the models in Tables 1 and 2, and 256 for the CUB-200 models in Table 3 and the ablations in App. I outside its last group. We use $D = 16$ on aircraft control and $D = 1$ on CarRacing and CUB-200, and the ablations in App. I vary D.

Each concept representation then passes through the capsule-style squash nonlinearity [Sabour et al., 2017],

$$\mathrm{squash}(v) = \frac{\|v\|^2}{1 + \|v\|^2} \cdot \frac{v}{\|v\|}, \tag{12}$$

applied to each of the J concepts separately. The squash bounds each representation's length to $[0, 1)$, and the concept classification loss trains this length to cluster near 0 and 1 (avoiding the mid ranges) as the concept's presence (Sec. 4.1).

E.2 Attention head h_ϕ

The head is a single-head attention layer with one learned query $q_r \in \mathbb{R}^D$ per output r. The concept representations $M \in \mathbb{R}^{J \times D}$ serve as both keys and values. The query, key, value, and output projections $W_Q, W_K, W_V, W_O \in \mathbb{R}^{D \times D}$ are all bias-free. For output r, the head computes the context

$$c_r = \sum_{j=1}^{J} \alpha_{rj} \, W_O W_V M_j, \qquad \alpha_{rj} = \mathrm{softmax}_j \left(\frac{(W_Q q_r)^\top W_K M_j}{\sqrt{D}} \right), \tag{13}$$

and the output

$$\hat{Y}_r = \tanh\big(w_r^\top c_r + b_r\big)\,, \tag{14}$$

where $w_r \in \mathbb{R}^D$ and $b_r \in \mathbb{R}$ are the weight and bias for output r. Since the projections are bias-free, a zero concept representation produces a zero key and a zero value. It therefore contributes no value to c_r, even though its attention weight stays positive (Prop. 1). On CUB-200, the tanh outputs of CCW-Net and the linear outputs of CBM are trained with the mean-squared error to the one-hot species label.

E.3 Per-concept invertible flow $\mathcal{F}_\psi$

Our chosen mediator model is a conditional normalizing flow with one independent flow per concept. Any modeling approach may be used. Each flow is conditioned on the labels of concept j's mutually exclusive set, the input, and a learned vector identifying concept j. For $D > 1$, each flow is a stack of 8 affine coupling layers whose conditioner networks have hidden width 256. For $D = 1$, a coupling layer cannot split the representation, so each flow is instead a stack of 8 conditional affine maps with hidden width 256. To draw a counterfactual for concept j, we flip the labels of its set and sample from the flow conditioned on them and the input (Eq. 1). The flow has its own optimizer (App. G) and is trained on the negative log-likelihood of the encoder's concept representations.

F Evaluation estimator

In short, we compare similar test inputs that differ in one concept and measure how much of the change in the model's output passes through each concept's representation (i.e., the fraction of change). The causal attribution estimator (Sec. 5.4) computes each reported fraction as follows. Let ℓ index concept label combinations, n_ℓ be the number of test inputs with concept label combination ℓ, and $\mathcal{C}(\ell)$ be the comparisons in which ℓ forms one of the two sets $\mathcal{A}$ and $\mathcal{B}$ (test inputs with original and swapped concept labels). Each comparison c swaps one active concept for a mutually exclusive alternative, so exactly two concept slots change labels. These two concept slots form one group, the swapped pair G_c^{sw}, and every other concept slot forms its own group.

Matching. We z-score the matching features $z \in \mathbb{R}^K$ (Sec. 5.4) over the test split. For a test input i and a set $\mathcal{S} \in \{\mathcal{A}, \mathcal{B}\}$, a match $i' \in \mathcal{S}$ is drawn with probability

$$p(i' \mid i) \propto \exp\big(-d_{ii'}^2/2\tau^2\big)\,\mathbb{1}[d_{ii'} \le \epsilon]\,\mathbb{1}[i' \neq i], \qquad d_{ii'} = \Big(\frac{1}{K}\sum_{k=1}^{K}(z_{ik} - z_{i'k})^2\Big)^{1/2}, \tag{15}$$

with bandwidth $\tau = 0.25$ and radius $\epsilon = 1.5$. When no input of $\mathcal{S}$ is within the radius, we take the nearest one. Each input receives 96 independently drawn matches for each group and set.

Comparisons. We form a comparison only when $|\mathcal{A}| \ge 100$ and $|\mathcal{B}| \ge 100$. For each concept label combination ℓ, we fix one random subset of $\min(n_\ell, 1000)$ test inputs, and these inputs serve as i in every comparison in $\mathcal{C}(\ell)$. Matches come from all test inputs of each set.

Group effects. $\text{IE}_G^{(c)}$ is Eq. (6) for comparison c, averaged over the inputs, the drawn matches, and all orderings of the groups. We use the Euclidean norm $\|\text{IE}_G^{(c)}\|$ of this averaged change in the model's output.

Concept slot effects. For concept label combination ℓ, let $\mathcal{C}_j(\ell) = \{c \in \mathcal{C}(\ell) : j \notin G_c^{\text{sw}}\}$ be the comparisons that keep concept slot j outside the swapped pair, and let $G_c(j)$ be concept slot j's group in comparison c. Concept slot j's effect is

$$\|\text{IE}_j\|^{(\ell)} = \begin{cases} \dfrac{1}{|\mathcal{C}_j(\ell)|}\displaystyle\sum_{c \in \mathcal{C}_j(\ell)} \big\|\text{IE}_{G_c(j)}^{(c)}\big\| & \text{if } \mathcal{C}_j(\ell) \neq \emptyset, \\ \dfrac{1}{|\mathcal{C}(\ell)|}\displaystyle\sum_{c \in \mathcal{C}(\ell)} \big\|\text{IE}_{G_c^{\text{sw}}}^{(c)}\big\| & \text{if } \mathcal{C}_j(\ell) = \emptyset \text{ and } c_j = 1. \end{cases} \tag{16}$$

In the second case, concept slot j is active and lies in the swapped pair of every comparison in $\mathcal{C}(\ell)$. In any other case, concept label combination ℓ is not scored. The fractions $\rho_j^{(\ell)}$, $\rho_{\text{active}}^{(\ell)}$, and $\rho_{\text{inactive}}^{(\ell)}$ then follow from Eq. (7).

Overall values. All Active and All Inactive weight the scored concept label combinations Λ that have an active concept by their test counts,

$$\bar{\rho}_{\text{active}} = \frac{\sum_{\ell \in \Lambda} n_\ell \, \rho_{\text{active}}^{(\ell)}}{\sum_{\ell \in \Lambda} n_\ell}, \qquad \bar{\rho}_{\text{inactive}} = 1 - \bar{\rho}_{\text{active}}, \tag{17}$$

and we report their means over ten seeds with two-sided 95% Student-t intervals.

Effect decomposition. The total effect, measured on the real matched inputs, equals the sum of the group effects plus a residual IE_μ (Eq. (10)), and this identity holds exactly. Here IE_μ is the difference between real inputs and hybrids whose groups come from independent matches.

G Training details

Algorithm 1 outlines CCW-Net training. The wrapper (g_θ, h_ϕ) and the flow $\mathcal{F}_\psi$ have separate optimizers. Training is conducted in three phases. During warmup, the wrapper learns the task and concept classification. During the intermission, the mediator model is fit to the encoder's concept representations. During alignment, the wrapper trains on the full objective (Sec. 4.4), while the mediator model supplies the counterfactual draws for the necessity loss. The mediator model is refit every E_r epochs during alignment so that it remains current with the encoder. Since the CUB-200 models train without the alignment losses they remain in the warmup phase throughout.

Algorithm 1 CCW-Net training. p_ψ denotes the flow's conditional density.

Require: Dataset $\mathcal{D} = \{(X_i, Y_i, c_i)\}$, frozen backbone f, warmup epochs E_w, total epochs E, refit interval E_r, draws per concept S

Initialize encoder g_θ and head h_ϕ
for epoch $= 1, \ldots, E_w$ **do** ▷ Warmup
  for batch $(X, Y, c) \in \mathcal{D}$ **do**
    $M \leftarrow g_\theta(f(X))$, $\hat{Y} \leftarrow h_\phi(M)$
    Update (θ, ϕ) on $\mathcal{L}_{\text{task}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}$
  end for
end for
Fit $\mathcal{F}_\psi$ on $\mathcal{L}_{\text{flow}} = -\mathbb{E}\big[\sum_j \log p_\psi(M_j \mid c_j, X)\big]$ with M held fixed ▷ Intermission
for epoch $= E_w + 1, \ldots, E$ **do** ▷ Alignment
  if E_r epochs have passed since the last fit **then**
    Refit $\mathcal{F}_\psi$ on the current encoder's concept representations
  end if
  for batch $(X, Y, c) \in \mathcal{D}$ **do**
    $M \leftarrow g_\theta(f(X))$, $\hat{Y} \leftarrow h_\phi(M)$
    Draw S counterfactuals $\tilde{M}_j \sim \mathcal{F}_\psi(\cdot \mid 1 - c_j, X)$ for every concept j ▷ no gradient to ψ
    Compute Δ^0, Δ_j^0, Δ_j^{cf}, and $\mathcal{L}_{\text{align}}$ (Sec. 4.3)
    Update (θ, ϕ) on $\mathcal{L}_{\text{wrap}} = \mathcal{L}_{\text{task}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{align}}$
  end for
end for

G.1 Loss weights

Loss weights for each domain are shown in Table 5. The task loss has weight 1 everywhere, and the four alignment losses share a common weight. The presence threshold is set as $\beta = 1/\sum_j c_j$ for equal share across each active concept. This gives $\beta = 0.5$ on aircraft control, where two concepts are active per input, and $\beta = 1$ on CarRacing, where one concept is active for any given input.

Table 5: Loss weights per domain. The CUB-200 models train without causal alignment losses.

Domain	λ_{cls}	λ_{nec}	λ_{suf}	λ_{ind}	λ_{pres}	β
Aircraft control	1.0	0.1	0.1	0.1	0.1	0.5
CarRacing	1.0	0.1	0.1	0.1	0.1	1.0
CUB-200, CCW-Net	1.0	—	—	—	—	—
CUB-200, CBM	0.01	—	—	—	—	—

G.2 Optimizers and learning rates

CCW-Net and the mediator model each use Adam with no weight decay, and the mediator model uses the wrapper's batch size (Table 6).

Table 6: Optimizer settings per domain. The CUB-200 models use no mediator model as they had no causal alignment losses active. App. I lists the batch size of each CarRacing ablation.

Domain	Wrapper learning rate	Flow learning rate	Batch size
Aircraft control	5×10^{-4}	1×10^{-3}	2048
CarRacing, Table 2	1×10^{-3}	1×10^{-3}	32
CarRacing, App. I canonical	1×10^{-3}	1×10^{-3}	128
CUB-200	5×10^{-4}	—	512

G.3 Training schedule

Table 7: Training schedule per domain. The mediator model is fit for 50 epochs at the start of alignment and refit every 10 alignment epochs to refresh with the current concept representations. Each alignment step uses $S = 8$ counterfactual draws per concept. The CUB-200 models train the task and concept losses for all 100 epochs.

Domain	Warmup epochs E_w	Total epochs E	Flow fit epochs	Refit interval E_r
Aircraft control	10	100	50	10
CarRacing	10	50	50	10
CUB-200	100	100	—	—

G.4 Data quantities

- Aircraft control: 500K samples for the main models. The data-quantity ablations use sets of 5K, 50K, and 5M samples.
- CarRacing: 15,102 samples from 20 episodes.
- CUB-200: 11,788 images, split into 72% train, 13% validation, and 15% test (8,516, 1,503, and 1,769 images).

H Task details

H.1 Aircraft control

In extended trail formation [United States Air Force, 2024, 2025], a chase aircraft maintains a volume of space behind a lead aircraft that flies dynamic maneuvers. The simulator contains a high-fidelity aircraft dynamics model with six degrees of freedom [Heidlauf et al., 2018]. For this experiment, the lead aircraft follows a scripted trajectory, and the chase aircraft is controlled by a reinforcement learning policy trained with proximal policy optimization [Schulman et al., 2017] using the aircraft environment and codebase of So and Fan [2023].

Backbone and action space. The frozen backbone policy maps the state-based observation X to a two-dimensional action Y of pitch and roll commands.

Concepts. $J = 4$ binary concepts drawn from operational pilot vocabulary [United States Air Force, 2024, 2025] providing real-world external validity to the concept set. No real-world concepts were dropped from the operational ontology for this task:

- *Lead pursuit*: the chase aircraft's nose points ahead of the lead aircraft.
- *Lag pursuit*: the chase aircraft's nose points behind the lead aircraft.
- *Climb*: the aircraft is ascending in altitude.
- *Dive*: the aircraft is descending in altitude.

The pursuit concept set (Lead, Lag) and the vertical concept set (Climb, Dive) are each mutually exclusive. Exactly one concept per set and two concepts overall are active per sample.

H.2 CarRacing

CarRacing-v1 [Brockman et al., 2016] is a top-down driving task with pixel inputs. The agent receives a 96×96 RGB image of the track and outputs continuous steering, throttle, and brake commands. We use the recorded latents, actions, and concept labels released with CW-Net [Kenny et al., 2026].

Backbone and action space. The frozen backbone is the convolutional policy from that release, trained end-to-end on CarRacing-v1. The action space is three-dimensional, with steering in $[-1, 1]$, throttle in $[0, 1]$, and brake in $[0, 1]$.

Concepts. $J = 3$ binary concepts, which the CW-Net authors labeled manually on over 15,000 examples [Kenny et al., 2026]:

- *Straight*: drive straight.
- *Left*: left turn.
- *Right*: right turn.

The three concepts are mutually exclusive and exhaustive, so exactly one concept is active per sample. The labels do not follow the agent's instantaneous steering command.

H.3 CUB-200 classification

CUB-200-2011 [Wah et al., 2011] is an image classification dataset of 200 bird species. We follow the concept bottleneck setup of Koh et al. [2020], in which 112 binary visual attributes, assigned per species, exist between the image and the species label.

Backbone and output space. The frozen backbone is Inception-v3, fine-tuned as the joint concept bottleneck model of Koh et al. [2020] on a species-stratified 80% of all 11,788 CUB-200 images. This fine-tuning set includes about 80% of our test images. The wrapper output is a 200-way species prediction.

Concepts. $J = 112$ binary visual attributes (e.g., *has yellow belly*, *has pointed beak*, *has streaked wing*). Each species has its own subset of attributes, so the number of active concepts varies by image, with a mean of about 7 over the test split.

Splits. We split the images into 72% train, 13% validation, and 15% test (8,516, 1,503, and 1,769 images), with a different split for each seed.

I Full ablation matrix

Tables 8 and 9 report every ablation on aircraft control and CarRacing. Outside the last group, every CCW-Net model uses an encoder of width 256 (App. E) and compares against the canonical model in the first row, and the baselines use their own architectures. The last group uses an encoder of width 64, the width of the models in Tables 1 and 2. All Active and All Inactive average ρ_{active} and ρ_{inactive} over the label combinations, weighted by each combination's number of test inputs (App. F). Loss

combinations name the alignment losses used, N (necessity), S (sufficiency), I (independence), and P (presence), and the canonical model uses all four. All values are means over ten seeds with two-sided 95% Student-t intervals, computed from the trained weights. App. F gives the estimator's settings.

I.1 Aircraft control

Table 8: Aircraft control: full ablation matrix. tMSE is the test action MSE, and concept accuracy is on the Lead/Lag and Climb/Dive concept sets (%). With a placebo concept added, the placebo carries $36.2 \pm 11.4\%$ of the effect at width 256, and $28.2 \pm 9.8\%$ with and $98.2 \pm 2.9\%$ without the alignment losses at width 64.

		Concept acc. (%)		Concept Effect (%)		DCI		
Model	tMSE	Lead/Lag	Climb/Dive	All Active ↑	All Inactive ↓	Disentanglement	Completeness	Informativeness
Ours								
CCW-Net (Ours, 500K samples)	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	97.3 ±1.1	2.7 ±1.1	0.999 ±0.001	0.748 ±0.050	0.979 ±0.000
CCW-Net (Ours, 5M samples)	0.120 ±0.000	97.5 ±0.0	99.7 ±0.0	98.3 ±0.9	1.7 ±0.9	1.000 ±0.000	0.780 ±0.043	0.986 ±0.000
CCW-Net (Ours, scalar $D = 1$)	0.412 ±0.003	96.2 ±0.1	99.4 ±0.0	69.9 ±11.2	30.1 ±11.2	0.998 ±0.001	0.896 ±0.065	0.978 ±0.001
No causal alignment losses								
CCW-Net ($D = 16$)	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	89.2 ±7.5	10.8 ±7.5	0.999 ±0.000	0.703 ±0.025	0.980 ±0.000
CCW-Net (scalar $D = 1$)	0.409 ±0.002	96.1 ±0.1	99.4 ±0.0	70.8 ±11.9	29.2 ±11.9	0.996 ±0.003	0.816 ±0.089	0.978 ±0.001
Architecture baselines								
CW-Net architecture, scalar + alignment	0.233 ±0.031	91.0 ±0.3	98.2 ±0.1	51.7 ±2.3	48.3 ±2.3	0.999 ±0.001	0.624 ±0.078	0.929 ±0.004
CW-Net (Kenny et al.)	0.122 ±0.002	91.2 ±0.2	98.2 ±0.1	50.0 ±6.2	50.0 ±6.2	0.998 ±0.001	0.719 ±0.064	0.947 ±0.001
Disentanglement baselines								
β-TCVAE ($D = 16$)	0.403 ±0.004	50.5 ±0.8	50.0 ±1.0	46.2 ±7.6	53.8 ±7.6	0.072 ±0.056	0.743 ±0.024	0.791 ±0.013
iVAE ($D = 16$)	0.401 ±0.002	52.5 ±8.8	49.1 ±7.6	47.6 ±2.4	52.4 ±2.4	1.000 ±0.000	0.891 ±0.051	1.000 ±0.000
Data quantity								
5K samples	0.378 ±0.012	91.7 ±0.3	98.0 ±0.4	77.2 ±8.4	22.8 ±8.4	0.981 ±0.010	0.718 ±0.056	0.954 ±0.009
50K samples	0.133 ±0.003	94.6 ±0.1	99.0 ±0.0	92.0 ±3.0	8.0 ±3.0	0.992 ±0.003	0.740 ±0.046	0.967 ±0.002
Label fraction								
10% labels	0.125 ±0.001	95.2 ±0.1	99.2 ±0.0	93.3 ±5.1	6.7 ±5.1	0.999 ±0.001	0.723 ±0.031	0.972 ±0.000
25% labels	0.124 ±0.001	95.8 ±0.1	99.4 ±0.0	97.0 ±1.4	3.0 ±1.4	0.998 ±0.001	0.722 ±0.056	0.976 ±0.000
50% labels	0.124 ±0.001	96.1 ±0.1	99.4 ±0.0	95.2 ±3.3	4.8 ±3.3	0.999 ±0.000	0.743 ±0.052	0.978 ±0.000
Label noise								
10% noise	0.121 ±0.000	96.2 ±0.1	99.4 ±0.0	71.6 ±8.0	28.4 ±8.0	1.000 ±0.000	0.773 ±0.031	0.978 ±0.000
20% noise	0.121 ±0.001	95.7 ±0.1	99.3 ±0.0	66.1 ±8.3	33.9 ±8.3	0.999 ±0.002	0.861 ±0.047	0.975 ±0.000
Loss combinations (without P)								
N	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	91.9 ±7.8	8.1 ±7.8	0.998 ±0.001	0.746 ±0.023	0.979 ±0.000
S	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	93.8 ±4.3	6.2 ±4.3	0.998 ±0.001	0.711 ±0.035	0.979 ±0.000
I	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	92.3 ±5.0	7.7 ±5.0	0.999 ±0.001	0.684 ±0.029	0.979 ±0.000
N+S	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	96.4 ±1.3	3.6 ±1.3	0.998 ±0.001	0.734 ±0.043	0.980 ±0.001

Table 8 continued

		Concept acc. (%)		Concept Effect (%)		DCI		
Model	tMSE	Lead/Lag	Climb/Dive	All Active ↑	All Inactive ↓	Disentanglement	Completeness	Informativeness
N+I	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	94.9 ±4.9	5.1 ±4.9	0.998 ±0.001	0.715 ±0.017	0.979 ±0.000
S+I	0.123 ±0.000	96.4 ±0.1	99.5 ±0.0	94.7 ±2.9	5.3 ±2.9	0.998 ±0.001	0.703 ±0.028	0.979 ±0.000
N+S+I	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	98.0 ±0.7	2.0 ±0.7	0.998 ±0.001	0.714 ±0.055	0.979 ±0.000
Loss combinations (with P)								
P	0.124 ±0.001	96.4 ±0.1	99.5 ±0.0	93.0 ±3.1	7.0 ±3.1	0.998 ±0.001	0.708 ±0.056	0.979 ±0.000
N+P	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	93.3 ±6.0	6.7 ±6.0	0.998 ±0.002	0.747 ±0.033	0.980 ±0.000
S+P	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	93.1 ±3.9	6.9 ±3.9	0.998 ±0.001	0.720 ±0.041	0.979 ±0.000
I+P	0.124 ±0.001	96.4 ±0.1	99.5 ±0.0	89.5 ±4.9	10.5 ±4.9	0.999 ±0.001	0.699 ±0.031	0.979 ±0.000
N+S+P	0.124 ±0.001	96.4 ±0.1	99.5 ±0.0	94.8 ±3.2	5.2 ±3.2	0.998 ±0.001	0.750 ±0.069	0.979 ±0.000
N+I+P	0.124 ±0.001	96.4 ±0.1	99.5 ±0.0	94.4 ±5.5	5.6 ±5.5	0.999 ±0.001	0.757 ±0.049	0.979 ±0.000
Concept ontology, merged concepts								
Merge Climb+Dive	0.122 ±0.000	96.4 ±0.1	55.6 ±0.0	97.4 ±1.1	2.6 ±1.1	0.823 ±0.095	0.573 ±0.047	0.842 ±0.007
Merge Lead+Lag	0.121 ±0.001	41.4 ±0.0	99.5 ±0.0	95.6 ±5.1	4.4 ±5.1	0.885 ±0.067	0.566 ±0.041	0.833 ±0.005
Concept ontology, removed concepts								
Remove Climb	0.124 ±0.001	96.4 ±0.1	44.4 ±0.0	93.9 ±1.5	6.1 ±1.5	0.998 ±0.001	0.764 ±0.042	0.979 ±0.000
Remove Dive	0.124 ±0.001	96.4 ±0.1	55.6 ±0.0	94.6 ±3.8	5.4 ±3.8	0.999 ±0.001	0.792 ±0.042	0.980 ±0.000
Remove Lag	0.123 ±0.001	41.4 ±0.0	99.4 ±0.0	79.9 ±11.4	20.1 ±11.4	0.999 ±0.001	0.750 ±0.039	0.980 ±0.000
Remove Lead	0.123 ±0.001	58.6 ±0.0	99.5 ±0.0	85.9 ±12.0	14.1 ±12.0	0.998 ±0.001	0.774 ±0.049	0.980 ±0.000
Placebo								
Placebo concept added	0.123 ±0.001	96.4 ±0.1	99.5 ±0.0	61.1 ±11.4	38.9 ±11.4	0.999 ±0.001	0.770 ±0.028	0.980 ±0.000
Encoder width 64								
CCW-Net (Ours), $D = 16$	0.125 ±0.001	95.7 ±0.1	99.3 ±0.0	94.5 ±2.2	5.5 ±2.2	0.999 ±0.001	0.719 ±0.041	0.975 ±0.001
CCW-Net (Ours), $D = 8$	0.125 ±0.001	95.6 ±0.1	99.3 ±0.0	93.1 ±5.2	6.9 ±5.2	0.999 ±0.000	0.717 ±0.053	0.974 ±0.001
10% label noise	0.121 ±0.001	95.3 ±0.1	99.2 ±0.0	70.4 ±5.5	29.6 ±5.5	1.000 ±0.000	0.762 ±0.031	0.973 ±0.000
10% label noise, no alignment	0.121 ±0.000	95.3 ±0.1	99.2 ±0.0	59.6 ±12.2	40.4 ±12.2	1.000 ±0.000	0.750 ±0.040	0.973 ±0.001
Placebo concept added	0.124 ±0.001	95.6 ±0.1	99.3 ±0.0	67.0 ±10.8	33.0 ±10.8	0.999 ±0.001	0.711 ±0.043	0.975 ±0.001
Placebo concept added, no alignment	0.121 ±0.001	95.7 ±0.1	99.4 ±0.0	0.9 ±1.8	99.1 ±1.8	0.999 ±0.001	0.685 ±0.020	0.975 ±0.001

I.2 CarRacing

Table 9: CarRacing: full ablation matrix. tMSE is the test action MSE, and concept accuracy is on the Straight/Left/Right concept set (%). Every group except the last uses batch size 128 for the canonical, $D = 16$ no-alignment, loss-combination, label-noise, and placebo rows and 32 for the other rows. The last group uses batch size 32. Every row with the presence loss uses $\beta = 1$ except CCW-Net (Ours, scalar $D = 1$) and the two CW-Net rows with alignment, which use $\beta = 0.5$. For the Remove rows, the averages cover the label combinations with an active concept, since the removed concept's own combination has none. § In the Merge rows, the estimator scores only the merged concept's label combination, because the remaining combination has no comparison that holds the merged concept fixed. With a placebo concept added, the placebo carries $34.6 \pm 6.8\%$ of the effect.

			Concept Effect (%)		DCI		
Model	tMSE	Concept acc. (%)	All Active ↑	All Inactive ↓	Disentanglement	Completeness	Informativeness
Ours							
CCW-Net (Ours, $\beta = 1.0$)	0.0025 ±0.0004	93.4 ±0.2	75.3 ±3.8	24.7 ±3.8	0.914 ±0.015	0.706 ±0.042	0.950 ±0.004
CCW-Net (Ours, scalar $D = 1$)	0.0420 ±0.0006	93.0 ±0.3	92.4 ±2.7	7.6 ±2.7	0.866 ±0.021	0.866 ±0.021	0.947 ±0.005
No causal alignment losses							
CCW-Net ($D = 16$)	0.0032 ±0.0004	93.4 ±0.3	63.5 ±9.0	36.5 ±9.0	0.913 ±0.011	0.680 ±0.034	0.950 ±0.003
CCW-Net (scalar $D = 1$)	0.0419 ±0.0006	93.0 ±0.4	90.0 ±5.5	10.0 ±5.5	0.870 ±0.012	0.871 ±0.012	0.948 ±0.003
Architecture baselines							
CW-Net architecture, scalar + alignment	0.0276 ±0.0089	89.8 ±1.3	37.2 ±6.7	62.8 ±6.7	0.085 ±0.061	0.560 ±0.083	0.928 ±0.009
CW-Net (Kenny et al.)	0.0015 ±0.0007	90.0 ±1.3	33.1 ±1.6	66.9 ±1.6	0.078 ±0.063	0.626 ±0.082	0.928 ±0.008
CW-Net architecture, $D = 16$ + alignment	0.0058 ±0.0015	92.3 ±0.5	52.2 ±2.9	47.8 ±2.9	0.763 ±0.024	0.710 ±0.021	0.938 ±0.005
CW-Net architecture, $D = 16$	0.0004 ±0.0001	92.4 ±0.4	43.3 ±2.4	56.7 ±2.4	0.772 ±0.031	0.678 ±0.034	0.938 ±0.003
Disentanglement baselines							
β-TCVAE ($D = 1$)	0.0290 ±0.0024	31.0 ±7.6	37.9 ±8.1	62.1 ±8.1	0.026 ±0.019	0.472 ±0.132	0.890 ±0.014
β-TCVAE ($D = 16$)	0.0293 ±0.0006	32.2 ±1.8	51.7 ±8.8	48.3 ±8.8	0.047 ±0.012	0.813 ±0.025	0.894 ±0.005
iVAE ($D = 1$)	0.0277 ±0.0009	32.4 ±13.0	39.0 ±7.0	61.0 ±7.0	0.022 ±0.018	0.354 ±0.079	0.897 ±0.009
iVAE ($D = 16$)	0.0191 ±0.0013	45.1 ±11.0	37.3 ±4.8	62.7 ±4.8	0.250 ±0.066	0.689 ±0.019	0.911 ±0.006
Label fraction							
10% labels	0.0026 ±0.0003	90.2 ±0.5	77.9 ±4.9	22.1 ±4.9	0.909 ±0.011	0.675 ±0.030	0.930 ±0.007
25% labels	0.0030 ±0.0005	91.2 ±0.4	75.6 ±7.3	24.4 ±7.3	0.910 ±0.012	0.684 ±0.046	0.936 ±0.007
50% labels	0.0032 ±0.0003	92.0 ±0.4	77.1 ±6.9	22.9 ±6.9	0.926 ±0.023	0.681 ±0.050	0.941 ±0.003
Label noise							
10% noise	0.0024 ±0.0002	92.3 ±0.3	61.8 ±9.3	38.2 ±9.3	0.923 ±0.009	0.753 ±0.049	0.944 ±0.004
20% noise	0.0023 ±0.0001	91.7 ±0.4	44.9 ±8.3	55.1 ±8.3	0.885 ±0.020	0.719 ±0.044	0.940 ±0.005
Loss combinations (without P)							
N	0.0031 ±0.0004	93.2 ±0.2	77.2 ±6.1	22.8 ±6.1	0.923 ±0.012	0.685 ±0.034	0.948 ±0.004

Table 9 continued

			Concept Effect (%)		DCI		
Model	tMSE	Concept acc. (%)	All Active ↑	All Inactive ↓	Disentanglement	Completeness	Informativeness
S	0.0025 ±0.0004	93.4 ±0.2	72.4 ±7.0	27.6 ±7.0	0.924 ±0.010	0.676 ±0.024	0.949 ±0.003
I	0.0031 ±0.0002	93.4 ±0.2	63.9 ±8.3	36.1 ±8.3	0.919 ±0.014	0.670 ±0.029	0.950 ±0.004
N+S	0.0025 ±0.0004	93.4 ±0.3	78.1 ±4.9	21.9 ±4.9	0.921 ±0.014	0.725 ±0.030	0.948 ±0.004
N+I	0.0032 ±0.0004	93.4 ±0.3	76.1 ±2.2	23.9 ±2.2	0.917 ±0.019	0.674 ±0.041	0.948 ±0.005
S+I	0.0026 ±0.0005	93.2 ±0.3	72.4 ±8.4	27.6 ±8.4	0.927 ±0.009	0.695 ±0.033	0.948 ±0.002
N+S+I	0.0026 ±0.0002	93.3 ±0.4	74.2 ±4.1	25.8 ±4.1	0.926 ±0.011	0.708 ±0.023	0.950 ±0.003
Loss combinations (with P)							
P	0.0033 ±0.0005	93.3 ±0.1	71.9 ±8.9	28.1 ±8.9	0.919 ±0.012	0.684 ±0.029	0.950 ±0.005
N+P	0.0030 ±0.0004	93.2 ±0.2	77.4 ±4.1	22.6 ±4.1	0.918 ±0.012	0.682 ±0.030	0.949 ±0.004
S+P	0.0026 ±0.0004	93.5 ±0.3	75.2 ±5.9	24.8 ±5.9	0.922 ±0.016	0.689 ±0.034	0.950 ±0.004
I+P	0.0031 ±0.0004	93.5 ±0.3	67.6 ±9.6	32.4 ±9.6	0.928 ±0.009	0.726 ±0.037	0.951 ±0.004
N+S+P	0.0026 ±0.0004	93.3 ±0.2	77.7 ±5.6	22.3 ±5.6	0.922 ±0.015	0.698 ±0.038	0.948 ±0.002
N+I+P	0.0031 ±0.0004	93.3 ±0.3	76.0 ±4.7	24.0 ±4.7	0.923 ±0.011	0.699 ±0.032	0.949 ±0.003
S+I+P	0.0025 ±0.0004	93.3 ±0.2	76.2 ±3.5	23.8 ±3.5	0.927 ±0.017	0.668 ±0.043	0.950 ±0.002
Concept ontology, merged concepts							
Merge Left+Right §	0.0023 ±0.0002	84.4 ±0.6	56.9 ±9.9	43.1 ±9.9	0.625 ±0.097	0.553 ±0.047	0.920 ±0.006
Merge Straight+Left §	0.0015 ±0.0003	72.1 ±0.2	95.7 ±2.5	4.3 ±2.5	0.629 ±0.045	0.481 ±0.027	0.842 ±0.011
Merge Straight+Right §	0.0019 ±0.0002	84.8 ±0.3	90.1 ±7.2	9.9 ±7.2	0.538 ±0.073	0.542 ±0.039	0.909 ±0.005
Concept ontology, removed concepts							
Remove Left	0.0036 ±0.0003	72.4 ±0.3	71.1 ±9.1	28.9 ±9.1	0.796 ±0.039	0.668 ±0.039	0.945 ±0.004
Remove Right	0.0049 ±0.0008	85.1 ±0.3	82.1 ±5.4	17.9 ±5.4	0.826 ±0.040	0.652 ±0.027	0.944 ±0.006
Remove Straight	0.0085 ±0.0013	35.0 ±0.1	65.8 ±10.3	34.2 ±10.3	0.637 ±0.114	0.688 ±0.023	0.948 ±0.004
Placebo							
Placebo concept added	0.0055 ±0.0008	93.2 ±0.2	51.3 ±7.9	48.7 ±7.9	0.935 ±0.012	0.701 ±0.034	0.949 ±0.004
Encoder width 64							
CCW-Net (Ours), $D = 16$	0.0031 ±0.0002	92.6 ±0.5	71.9 ±5.7	28.1 ±5.7	0.921 ±0.011	0.686 ±0.031	0.945 ±0.005
CCW-Net (Ours), $D = 8$	0.0037 ±0.0004	92.7 ±0.4	67.8 ±6.1	32.2 ±6.1	0.880 ±0.010	0.673 ±0.039	0.946 ±0.004